IVF/NSGAII: In Vitro Fertilization Method Coupled to NSGAII

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***Abstract*—The In Vitro Fertilization Genetic Algorithm - IVF/GA is a promising algorithm applicable to mono-objective problems, especially for complex and multimodal ones. Due to the balance between the exploration and exploitation of the IVF/GA, and thus its abilities to avoid the local optimum, we speculate that the IVF can also improve the Multi-Objective Evolutionary Algorithms (MOEA). So, this paper proposes the IVF/NSGAII, which is the IVF coupled to NSGAII. We evaluate the proposal's improvement comparing the canonical NSGAII versus IVF/NSGAII, applied to Zitzler-Deb-Thiele (ZDT) functions. The results show the IVF/NSGAII outperformed the canonical version on Inverted Generational Distance (IGD) metric, mainly in the complex functions, e.g. ZDT 4, 5 and 6. Therefore, we conclude that IVF improved the NSGAII and also is a promising approach to support MOEA's.**

Keywords*—*In Vitro Fertilization Genetic Algorithm, NSGAII, Multiobjective Optimization, MOEA.

I. Introduction

There are many problems that require simultaneous optimization of two or more conflicting objectives in the real world. This class of problem is called multi-objective optimization (MOP). In MOPs, the gain of one objective implies the loss of the other. Thus, we do not have a single solution, but a set of optimal solutions (also known as Pareto-optimal solutions), which indicate the best compromise between the different objectives considered. [1]

Different optimization techniques can be used to obtain the pareto-optimal set. However, Evolutionary Algorithms (EAs) have shown good performance in the treatment of MOPs [2], although they do not guarantee the global optimum [3]. In addition to the multimodal characteristic, complex problems may occur. Considering the objectives space, different geometric shapes of the Pareto frontier, such as discontinuous shapes, can cause unsatisfactory performance of the EAs. These characteristics make convergence and exploration difficult in solution search space.

To obtain good results, it is important that such algorithms maintain their capability for convergence and diversification, which is not always achieved. In regard to these difficulties, the work of [3] is relevant, in which the In Vitro Fertilization Genetic Algorithm (IVF/GA) is presented in the form of a Parallel Auxiliary Algorithm, aiming to assist EAs with good individuals, using mining information present in the population of the aided Evolutionary Algorithm. IVF/GA manipulates a GA's subpopulation, simulating the generation of "test tube babies" by In Vitro Fertilization and then deploys the best individual on the aided algorithm. The promising previous results [3] and [4], including in multimodal problems [3], enable the algorithm to be tested and compared, with benchmarks in the multiobjective context.

In this paper, In Vitro Fertilization NSGAII (IVF/NSGAII) is proposed, adapting and coupling the IVF in NSGAII to treat multiobjective problems. The IVF is expected to assist the NSGAII in spreading the solutions over the border, in addition to improving convergence. In order to evaluate the contribution of IVF/NSGAII in the treatment of multiobjective problems, the proposal is validated in experiments with six functions of the ZDT benchmark [5], which explore different and complex Pareto front geometries in the objective space (convex, non-convex, discontinuous, multimodal, among others).

Thus, the main contributions of this work are: (1) The adaptation of IVF to multiobjective problems; (2) The IVF/NSGAII proposal to improve NSGAII performance on complex problems; and (3) An experimental analysis of the proposal in multiobjective and complex problems, with convex, non-convex, discontinuous or multimodal boundaries.

The other sections of the article are divided as follows: Section II discusses the concepts used in this work, such as NSGA-II and In Vitro Fertilization Genetic Algorithm; Section III presents the IVF/NSGA-II proposal; Section IV describes the experiments, benchmarks and parameters used in the validation of the proposal; Section V shows and discusses the results, and finally Section VI presents the conclusions and future work.

II. Concepts

A. Nondominated Sorting Genetic Algorithm II (NSGA-II)

NSGA-II is one of the most cited algorithms in the literature of multiobjective optimization using evolutionary algorithms. It was proposed by [1]. It uses elitism and orders its population based on non-dominance ranking. An individual's ranking number indicates how many solutions dominate him. Thus, Ranking 0 solutions are not dominated by any other solution. For the maintenance of diversity, NSGA-II includes a crowding distance attribute, calculated for each individual (solution), which identifies how close the solution is to its neighborhood. High values for this attribute may indicate a better diversity for the solutions.

The steps of NSGAII can be described as: (1) Create random initial population P of size N; (2) Evaluate each solution of the population; (3) Select parent by round; (4) Recombine parents to obtain two descending solutions, with a given probability for crossing; (5) Apply mutation in the new obtained solutions with a given probability for mutation; (6) Evaluate each new obtained solution; (7) Add population of descendants to the population of parents; (8) Sort new population by rank of dominance, in ascending order (the lower the rank, the better), and by crowding distance in decreasing order (the higher the crowding distance, the better); (9) Maintain in the population only the best N individuals and remove the remaining solutions; and (10) repeat from item 3 until stop criterion is reached.

B. In Vitro Fertilization Genetic Algorithm - IVF/GA

Proposed in [3], the In Vitro Fertilization Genetic Algorithm (IVF/GA) is an auxiliary algorithm with parallel flow, contributing to the maintenance of convergence and exploration in the search space. The In Vitro Fertilization module recombines chromosomes from part of the GA population to mine genetic information present in individuals. With this, it aims to identify, use and recombine good genetic material (building blocks), simulating the generation of “test tube babies” genetically manipulated and evaluated in the in vitro fertilization process.

As demonstrated in [3], the algorithm has the potential to be coupled to other evolutionary algorithms, since it is modular (In Vitro Fertilization Module, IVFm) and executes in a parallel flow to its host algorithm (Figure 1).

In Vitro contains the following steps: Collection, Genetic Manipulation and Transfer. The Collection stage is the extraction of part of the population (nuIndiv) that will be submitted to genetic manipulations for information mining and production of new solutions.

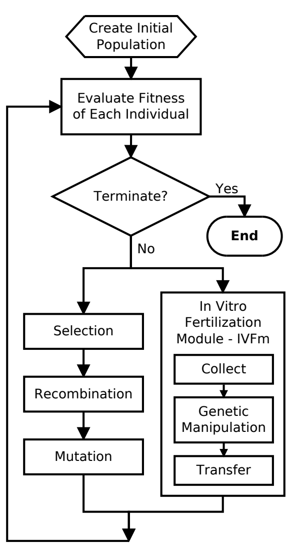


Fig. 1. GA with the IVFm

The Genetic Manipulation (GM) stage is divided into two parts. The first one subjects the collected population to the treatment of exploratory operators (Exploratory Assisted Recombination - EAR), either altering part of genetic material addressed (EAR-P and EAR-PA), or altering all the genetic material in the individuals (EAR-N and EAR-T), see Figure 2. One can avoid these genetic material modifications and uses

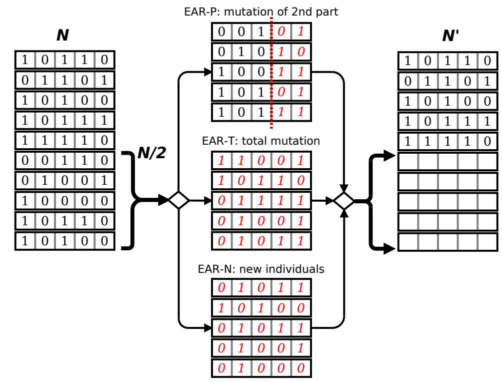


Fig. 2. Example of the first part of the Genetic Manipulation stage. The AR operator does not participate in this part.

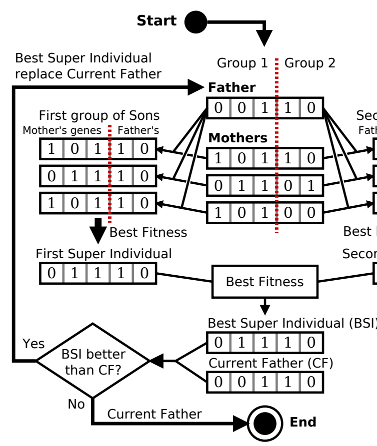


Fig. 3. Example of the second part of the Genetic Manipulation stage.

the original information (AR). The Assisted Recombination (AR) operator, without EAR, is used to better exploit the original genetic information of the collected sub-population. Table I shows a description of the different assisted recombination operators: AR, EAR-N, EAR-T, EAR-P and, finally, EAR-PA. In the second part of the GM, we subject the resulted sub-population from the first part to the genetic recombination operation, assigning a single individual as “Father” and the others as “Mothers”, according to Figure 3.

TABLE I. IN VITRO ASSISTED RECOMBINATION OPERATORS

Operator	Description
AR	It uses information from the individuals collected from the AG population, without genetic alterations. Provides speed in operation.
EAR-N	It replaces half of the collected mothers with new random individuals. Extend the exploration in the search space.
EAR-T	Mutates the entire chromosome from half of the collected mothers.
EAR-P	Mutates part of the chromosome from half of the collected mothers.
EAR-PA	Mutates part of the chromosome of some individuals from half of collected mothers.

The AR only applies the second part of the Genetic Manipulation stage (Figure 3). On the other hand, the EAR-P disturbs part of the genetic material of the individuals, defined by parameter. The EAR-T disturbs all the genetic material. And finally, EAR-N creates new individuals to form the subpopulation that will be the genetic material for the recombination part.

Finally, the Transfer stage inserts the "super individual" in the population of new solutions of the host algorithm, so that they are submitted to the process of ordering and cutting of this last algorithm and then generating its new population.

This process resembles, in part, the strategy of using an external population, as in SPEA and SPEA-2 algorithms [6]. However, in the case of IVF/GA, this "super-individual" external population comes from the mining of genetic information from good individuals extracted from the original population itself, or new individuals from the exploratory operator.

III. APPROACH

The work [3] which proposed In Vitro Fertilization Genetic Algorithm, and [4], indicates as suggestions for future work, its adaptation to the multiobjective problems, as well as the coupling to other Evolutionary Algorithms. Thus, we focus on adapting the IVF to NSGAII (IVF/NSGAII), as presented in this section.

The IVF/NSGAII, as well as the IVF/GA has three stages: Collection, Genetic Manipulation and Transfer. Below there are descriptions of each of the phases of the proposal.

A. Collect

Single-objective IVF/GA collection is usually performed from the choice of the best individuals in the GA population, sorted by the value of the single function to be optimized. However, in the multiobjective approach, the collection needs to take into account another strategy, since there is more than one objective function. The best individuals, from the point of view of a function, do not indicate the best individuals from the point of view of other functions of the problem. Thus, the collection strategy may be associated with the ordering strategy of the host algorithm, or may use any strategy in the MOEA literature. In the case of the IVF/NSGAII proposal, the ordering by non-dominance and by crowding distance is used.

Once the collection strategy has been defined, it is necessary to define the form of choice of the individual "Father", usually the best of the collection, and the "Mothers" of the other individuals. These labels are important for the next steps of the algorithm.

In IVF/GA, the best individual is one who has the best fitness. But in multiobjective, there is not a single better individual among those distributed on the same frontier. Therefore, in order to maintain the affinity with the NSGAII host algorithm, we consider that the best individuals are those sorted by ranking of non-dominance (ascending order), and by "crowding distance" (descending order). Figure 4 shows the flowchart of the Collect stage.

B. Genetic Manipulation

The Genetic Manipulation has two parts, the first one of perturbation of some individuals and the second of recombination. In IVF/NSGAII, the first part needs some adaptations. Exploratory assisted recombination operators, for example the EAR-P, mutate part of the chromosomes selected for the perturbation. In this multiobjective approach, and for coupling

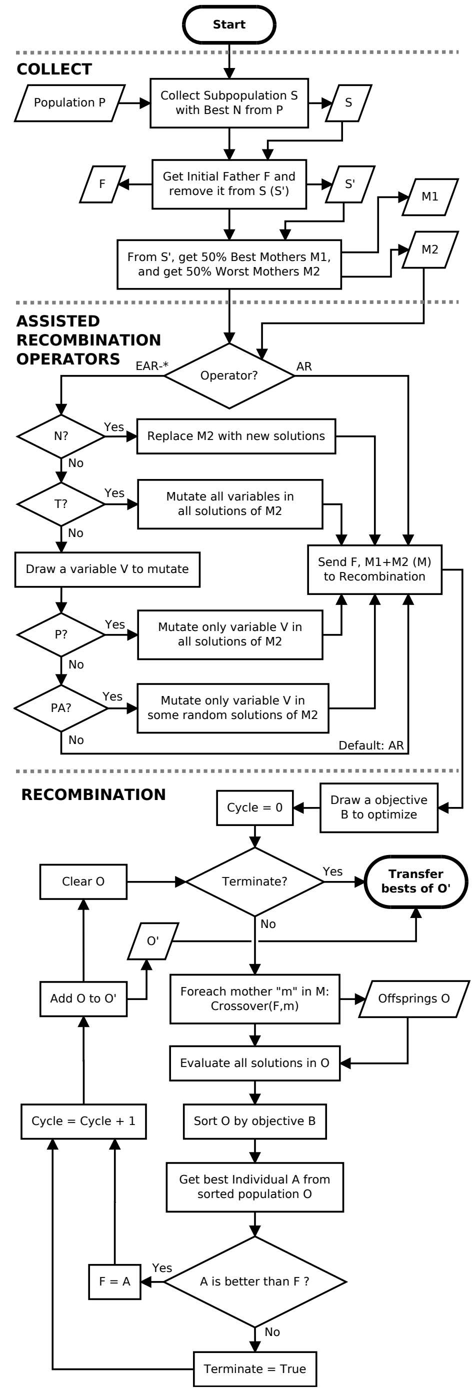


Fig. 4. IVF/NSGAII flowchart

in EAs with real genotypic representation, variables will be used as parts to be mutated. Thus, a variable will be drawn to suffer the perturbation, considering the probabilities, defined as parameters, of each variable to be chosen.

In the second part of the Genetic Manipulation, the IVF/NSGAII differs from the IVF/GA by changing the form of comparison between the "Father" and the offspring that were generated, for a new cycle. In IVF/GA, this comparison is made based on the fitness of the solutions. Thus, the best between the "Father" and the "Sons" is chosen as the new "Father" for the new cycle. In the IVF/NSGAII, the comparison can be made based on a current goal settled at the beginning of the Genetic Manipulation stage, which allows the intensification of different objectives at each execution of the In Vitro Fertilization module. This objective intensification alternation minimizes the selective pressure and thus premature convergence.

In addition to the comparison by objectives, it is also possible to use any strategy from the MOEA literature, for example, the ordering by non-dominance and by crowding distance. In order to avoid excessive consumption of evaluations, the IVF/NSGAII has a parameter to limit the maximum number of internal cycles of the genetic recombination part.

C. Transfer

The IVF/GA Transfer operator can be implemented in different ways. The most widely used is transferring to the host population a single super individual. However, one can also transfer a set of super individuals produced in the Genetic Manipulation stage. Super individuals with good results in one goal may be dominated by other solutions available in the EA host population. Other solutions with lower values on that same goal might not be dominated. Thus, sending only the best super-individual of a single goal, could cause its immediate disposal by NSGAII, since it would probably be dominated by another solution in some other objective. In addition, there is a possibility of losing good genetic material from other super individuals that are not dominated by the initial parent solution.

The computational cost, measured as the number of solution evaluations, is an important factor in optimization. Discarding reasonable solutions, at least to one objective, that have already been evaluated by IVF represents a loss in cost and probably in effectiveness. Thus, we recommend that the IVF/NSGAII transfer more than one super individual to the host MOEA population.

Finally, the IVF/NSGAII has an execution rate of the IVF module. Thus only a few generations of NSGAII will be affected by the implementation of IVF. This avoids excessive consumption of evaluations and enables the host EA to properly develop its own evolutionary strategy.

IV. Experiments

In this section, the benchmarks, algorithms and parameters used to validate the proposal of this work are presented. The IVF/NSGAII was tested on all six ZDT multi-objective benchmarking problems [5]. The following sections give more details about the tests performed.

A. ZDT Test Suite

For the evaluation of the IVF/NSGAII, the six problems of the multi-objective benchmark ZDT [5] were used, ZDT1 to ZDT6. This benchmark is well-know and widely used to assess the performance of algorithms for MOP [7]. It has some problems with real domain variables and one problem with binary domain variables (ZDT5). Their problems are complex and explore different Pareto-front geometries, such as: convex Pareto-front (ZDT1), non-convex (ZDT2), convex and disconnected (ZDT3), non-convex and multimodal (ZDT4 contains 21^9 local Pareto-fronts), non-convex with binary encoding (ZDT5) and non-uniformly distributed solutions over Pareto-front (ZDT6).

Based on the CEC'09 MOEA Competition [8], we use the Inverted Generational Distance (IGD) metric to evaluate our proposal.

B. Metric IGD

Let S be a Pareto set approximation and P a set of uniformly distributed points from the Pareto set. Let $\delta(x, y)$ be the Euclidean distance of points x and y in the objective space. The inverse generational distance metric is defined in Equation 1. Thus, if some points of the set P are not covered by the Pareto set approximation, the approximation is penalized.

$$IGD(S,P) = \frac{1}{|P|}\sum_{x\in P}\min_{y\in S}\delta(x,y) \tag{1}$$

C. Parametrization

Each ZDT problem was executed 30 times for each algorithm, with a maximum limit of 25,000 evaluations per seed. The NSGAII parameters were based on [1] and [9] experiments. This values are: number of individuals = 100, crossover (cx) operator = SBX (Simulated Binary Crossover), cx probability (pc) = 0.9, SBX distribution (η_c) = 15, mutation operator = PM (Polynomial Mutation), mutation probability (pm) = 0.01 (1/populationSize) and PM distribution (η_m) = 20. These values were used both in the canonical version (NSGAII) and in the version presented in this work (IVF/NSGAII).

In addition to the canonical parameters, the IVF/NSGAII has other parameters that were evaluated as described in Table II. With this, it is possible to evaluate the contribution of each operator and other parameters of the IVF for different properties of the problems. To maintain uniformity with the host algorithm, In Vitro genetic manipulation stage used the operators PM, for mutation of part of "mothers", and SBX, for genetic recombinations with "current father".

TABLE II. IVF/NSGAII PARAMETERS

Parameter	Description	Values
IVFop	Operator of the Genetic Manipulation Stage of In Vitro Fertilization module	AR, EAR-* (N, T, P, PA)
IVFm rate (R)	Probability of IVF module to be run in a generation of EA	0.1 to 1.0, step 0.1
Collect Size (C)	Number of individuals obtained in the Collect stage	5, 7, 9, 11, 13, 21, 31, 41
Max Cycles (M)	Maximum number of internal In Vitro recombination cycles	2 to 6, step 1

For each set of 30 executions (seeds) of each algorithm for each problem, IGD's measures are calculated: mean, standard deviation, median, minimum and maximum. These results were compared using the Wilcoxon rank sum test, with .05 significance level.

V. Results

In this section, the results of the aforementioned experiments are presented. Tables III, IV, V, VI, VII and VIII contain the algorithms' results for IGD metric. Only the seven best IVF/NSGAII configurations are listed, facilitating the understanding. IVF/NSGAII is tested with different parameters: IVFm operator (IVFop), Rate of execution of IVFm (R), Collect Size (C) - the size of subset of population to be manipulated in the IVF; and Max Cycles (M) - maximum cycling within the IVF.

A. ZDT1

Table III demonstrates that NSGAII already performs well on this problem, once NSGAII front is very similar to the Pareto-optimal front. Consequently, the IVF/NSGAII algorithms are not significantly best, despite its better average values. About the IVF/NSGAII parameters, the EAR-T operator obtained the highest number of best solutions, followed by AR and EAR-PA. The best IVFm rates (R) were 0.5 and 0.6, showing that few IVF interventions are sufficient for this problem. About Collect Size, the best results were obtained with only 5% and 7% of the NSGAII population. Finally, the best M were from 2 to 6.

TABLE III. Results obtained for problem ZDT1. Low IGD is better and the results are in descending order

Algorithm	IVF op	Inverted Generational Distance (IGD)					IVF conf.		
		Average	Std.Dv.	Median	Best	Worst	R	C	M
NSGAII		0.00484	0.00021	0.00485	0.0045	0.0053			
IVF/NSGAII	EAR-T	0.00480	0.00015	0.00482	0.0044	0.0051	0.1	5	3
IVF/NSGAII	EAR-T	0.00480	0.00016	0.00476	0.0045	0.0051	0.5	9	2
IVF/NSGAII	EAR-T	0.00480	0.00020	0.00479	0.0045	0.0054	0.2	13	4
IVF/NSGAII	EAR-PA	0.00479	0.00017	0.00475	0.0045	0.0055	0.5	5	2
IVF/NSGAII	AR	0.00479	0.00018	0.00476	0.0045	0.0052	0.2	7	3
IVF/NSGAII	AR	0.00476	0.00014	0.00475	0.0045	0.0051	0.6	5	6
IVF/NSGAII	EAR-T	0.00475	0.00020	0.00472	0.0044	0.0052	0.5	7	2

B. ZDT2

NSGAII also performs well for ZDT2 problem, similar to ZDT1 problem (Table IV). The NSGAII front is close to Pareto-optimal for this problem as well. The IVF/NSGAII maintains good results and also has superior value for some configurations. About the IVF/NSGAII parameters, the EAR-T has the highest number of best solutions, followed by AR, EAR-PA and EAR-P. The best IVFm rates (R) were 0.1 and 0.2, showing that few IVF interventions are sufficient for this problem. About Collect Size, the best results were obtained with only 5% and 7% of the NSGAII population. Finally, the best M were from 2 to 6.

TABLE IV. Results obtained for problem ZDT2.Low IGD is better and the results are in descending order

Algorithm	IVF op	Inverted Generational Distance (IGD)					IVF conf.		
		Average	Std.Dv.	Median	Best	Worst	R	C	M
IVF/NSGAII	EAR-T	0.00488	0.00017	0.00489	0.0046	0.0054	0.5	7	6
IVF/NSGAII	EAR-T	0.00488	0.00019	0.00490	0.0045	0.0054	0.2	7	6
IVF/NSGAII	EAR-T	0.00487	0.00020	0.00478	0.0045	0.0055	0.2	7	2
NSGAII		0.00487	0.00016	0.00485	0.0046	0.0053			
IVF/NSGAII	EAR-T	0.00487	0.00019	0.00483	0.0046	0.0053	1.0	5	2
IVF/NSGAII	EAR-PA	0.00486	0.00016	0.00488	0.0046	0.0051	0.2	5	3
IVF/NSGAII	EAR-P	0.00486	0.00021	0.00480	0.0045	0.0057	0.2	5	2
IVF/NSGAII	AR	0.00485	0.00017	0.00484	0.0045	0.0053	0.1	5	3

C. ZDT3

NSGAII presents good results for ZDT3 problem, as well as in ZDT1 and ZDT2 problems. The low standard deviation also proves robustness (Table V). The NSGAII front is close to Pareto-optimal for this problem as well. The IVF/NSGAII do not degenerated the good genetic material and thus obtained higher values for some configurations, even for disconnected scenario. The EAR-T operator obtained the greatest number of best solutions, followed by the AR. About the IVFm rate (R), the best results were 0.2 and 0.5, showing that few IVF interventions are sufficient for this problem as well. About Collect Size, the best results were obtained with 5%, 7% and 11% of the NSGAII population. Finally, the best M values were 5 and 6.

TABLE V. Results obtained for problem ZDT3. Low IGD is better and the results are in descending order

Algorithm	IVF op	Inverted Generational Distance (IGD)					IVF conf.		
		Average	Std.Dv.	Median	Best	Worst	R	C	M
NSGAII		0.00337	0.00012	0.00334	0.0032	0.0036			
IVF/NSGAII	EAR-T	0.00335	0.00012	0.00336	0.0032	0.0037	0.8	5	3
IVF/NSGAII	EAR-T	0.00335	0.00010	0.00336	0.0032	0.0036	0.9	5	5
IVF/NSGAII	EAR-T	0.00335	0.00013	0.00332	0.0031	0.0037	0.2	9	5
IVF/NSGAII	EAR-T	0.00335	0.00013	0.00334	0.0031	0.0037	0.2	9	2
IVF/NSGAII	AR	0.00335	0.00011	0.00334	0.0030	0.0035	0.5	5	6
IVF/NSGAII	EAR-T	0.00334	0.00015	0.00333	0.0031	0.0038	0.2	11	5
IVF/NSGAII	EAR-T	0.00332	0.00013	0.00327	0.0031	0.0037	0.2	7	6

D. ZDT4

ZDT4 has 21^9 local Pareto-fronts, which tests the ability of algorithms to deal with the multimodality. NSGAII has convergence issue for this problem and thus its fronts is not close to the Pareto-optimal front. On the other hand, the IVF/NSGAII obtains very expressive results due to its ability to escape of the local optimal. Figure 5 shows the fronts of IVF/NSGAII more close to Pareto-optimal front than NSGAII ones. Therefore, the contribution of the IVFm auxiliary strategy to the NSGAII host algorithm is substantial.

TABLE VI. Results for ZDT4. Low IGD is better and the results are in descending order

Algorithm	IVF op	Inverted Generational Distance (IGD)					IVF conf.		
		Average	Std.Dv.	Median	Best	Worst	R	C	M
NSGAII		0.49453	0.25190	0.52568	0.1254	1.4317			
IVF/NSGAII	EAR-N	0.31121	0.15349	0.31975	0.0045	0.6662	1.0	31	3
IVF/NSGAII	EAR-N	0.31097	0.10781	0.25637	0.0048	0.6664	1.0	41	5
IVF/NSGAII	EAR-N	0.30989	0.16265	0.25585	0.0047	0.6663	0.9	41	4
IVF/NSGAII	EAR-N	0.30756	0.16021	0.32632	0.0046	0.5265	1.0	31	5
IVF/NSGAII	EAR-N	0.30349	0.14926	0.25595	0.0049	0.6669	0.5	41	6
IVF/NSGAII	EAR-N	0.29783	0.12568	0.25995	0.1254	0.5264	0.6	31	6
IVF/NSGAII	EAR-N	0.26885	0.13773	0.25753	0.0046	0.5266	0.9	41	5

We highlight the EAR-N operator which obtained the seven best solutions (Table VI). The best IVFm rates (R) were 0.9 and 0.6, showing that more IVF interventions are necessary for this multimodal problem. About Collect Size, the best results were obtained with 41% and 31% of the NSGAII population, demonstrating the dependence for more genetic material of the IVF to combine good solutions with new ones produced by EAR-N. Finally, the best M were 5 and 6 demonstrating that more cycles improve the efficacy.

E. ZDT5

This problem uses binary encoding. Table VII shows that IVFm also outperformed the NSGAII and it is able to also lead with binary representation. The EAR-PA operator obtained the greatest number of best solutions, followed by the EAR-P, EAR-N and EAR-T. About the IVFm rate (R), the best results were 0.6 and 1.0, showing that many IVF interventions are necessary for this problem. About Collect Size, the best results were obtained with 5%, 7% and 9% of the NSGAII population. Finally, the best M values were 3 and 4.

TABLE VII. RESULTS OBTAINED FOR PROBLEM ZDT5. LOW IGD IS BETTER AND THE RESULTS ARE IN DESCENDING ORDER

Algorithm	IVF op	Inverted Generational Distance (IGD)					IVF conf.		
		Average	Std.Dv.	Median	Best	Worst	R	C	M
NSGAII		0.02111	0.01025	0.01850	0.0034	0.0473			
IVF/NSGAII	EAR-PA	0.01459	0.00949	0.01251	0.0008	0.0322	0.5	9	4
IVF/NSGAII	EAR-PA	0.01441	0.00972	0.01139	0.0020	0.0386	0.9	5	4
IVF/NSGAII	EAR-T	0.01416	0.01004	0.01152	0.0007	0.0465	0.5	5	5
IVF/NSGAII	EAR-P	0.01412	0.00909	0.01204	0.0019	0.0371	1.0	5	6
IVF/NSGAII	EAR-N	0.01399	0.00831	0.01257	0.0007	0.0325	0.5	7	6
IVF/NSGAII	EAR-PA	0.01357	0.00862	0.01195	0.0008	0.0324	1.0	5	3
IVF/NSGAII	EAR-P	0.01303	0.00873	0.01200	0.0007	0.0350	0.6	5	4

F. ZDT6

The Pareto-optimal solutions of the ZDT6 problem are not uniformly distributed over the Pareto-front, in the objective space. Furthermore, the density of solutions is sparse when close to the Pareto-front. Table VIII shows that IVF/NSGAII outperformed the NSGAII. The EAR-PA operator obtained the greatest number of best solutions, followed by the AR. About the IVFm rate (R), the best results were 0.2 and 0.5, showing that few IVF interventions are enough for this problem. About Collect Size, the best results were obtained with 5%, 7% and 9% of the NSGAII population. Finally, the best M value is 6.

TABLE VIII. RESULTS OBTAINED FOR PROBLEM ZDT6. LOW IGD IS BETTER AND THE RESULTS ARE IN DESCENDING ORDER

Algorithm	IVF op	Inverted Generational Distance (IGD)					IVF conf.		
		Average	Std.Dv.	Median	Best	Worst	R	C	M
NSGAII		0.01426	0.00234	0.01445	0.0089	0.0188			
IVF/NSGAII	EAR-PA	0.01329	0.00244	0.01265	0.0080	0.0177	0.2	7	3
IVF/NSGAII	AR	0.01329	0.00237	0.01313	0.0087	0.0185	0.5	7	6
IVF/NSGAII	AR	0.01327	0.00186	0.01336	0.0094	0.0177	0.5	7	3
IVF/NSGAII	EAR-PA	0.01315	0.00203	0.01331	0.0098	0.0190	0.2	7	2
IVF/NSGAII	EAR-PA	0.01303	0.00150	0.01322	0.0100	0.0160	0.5	7	6
IVF/NSGAII	EAR-PA	0.01302	0.00185	0.01308	0.0097	0.0163	0.2	9	6
IVF/NSGAII	AR	0.01287	0.00223	0.01348	0.0084	0.0180	0.5	5	6

VI. CONCLUSION

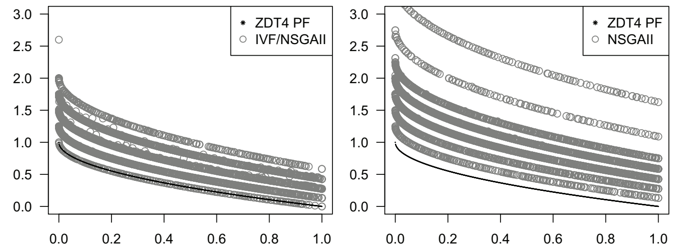
This work proposed an adaptation of the IVF/GA to deal with the multiobjective problems. The proposal is the IVF module coupled to NSGAII: IVF/NSGAII. The widely used ZDT problems were used to validate the approach.

The results show that the IVF/NSGAII can improve the NSGAII (Fig. 6), especially for a complex and multimodal problem as ZDT4 (Fig. 5). The results also show that each operator has a better suitable problem. Thus, we conclude the IVF/NSGAII significantly outperformed (p-value ≤ 0.05) the canonical NSGAII in the ZDT4, ZDT5 and ZDT6 problems (Table IX).

As future works, we will try different multiobjective benchmarks, like the Deb-Thiele-Laummans-Zitzler (DTLZ) problems, and others restrictive problems of CEC'09 MOEA Competition [8]. Finally, we will apply the auto-adaptation methods to the IVFm parameters.

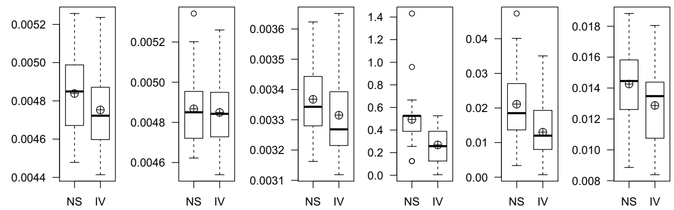
TABLE IX. BEST IVF/NSGAII OPERATOR VERSUS NSGAII - WILCOXON RANK SUM TEST P-VALUES

MOP:	ZDT1	ZDT2	ZDT3	ZDT4	ZDT5	ZDT6
p-value:	0.1058	0.7412	0.0514	0.001366	0.003096	0.01833



(a) IVF/NSGAII (b) NSGAII

Fig. 5. Algorithms' Pareto-fronts for highly multimodal ZDT4 problem. (a) 30 IVF/NSGAII fronts and (b) 30 NSGAII fronts. Black points are ZDT4 PF.



(a) ZDT1 (b) ZDT2 (c) ZDT3 (d) ZDT4 (e) ZDT5 (f) ZDT6

Fig. 6. Boxplots comparing IGDs': NSGAII (NS) and IVF/NSGAII (IV) (min, 1st quartile, median, mean (circle plus), 3rd quartile, max and outliers)

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